

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Engineering

L-4/T-1 CSE 409: Computer Graphics

Time: 25 minutes**Marks: 20**Student Name: Md. Ashrafur Rahman KhanStudent ID: 1905005

1. a) What happens if we shoot rays from light sources instead of the eye? (2)

Answer: It is very inefficient since most of the rays will not reach the eye (camera) and hence will not contribute to the final image.

- b) For a 3840×2160 resolution image, how many rays should be cast (at least) for a ray-traced image? (2)

Answer: At least one ray per pixel, so $3840 \times 2160 = 8,294,400$ rays.

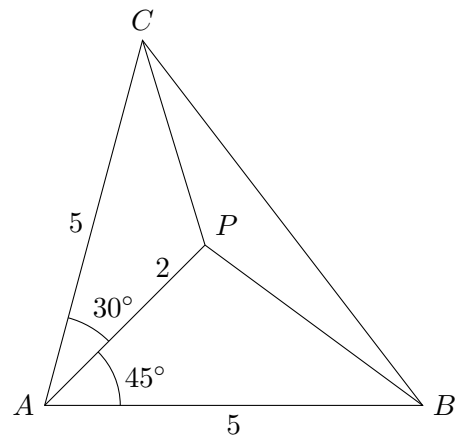
- c) During calculations, how do we ensure that we take the closest intersection? (2)

Answer: We keep track of the minimum positive distance (t value) from the ray origin to the intersection points and update it whenever we find a closer intersection.

2. a) In the triangle ABC what is the barycentric coordinates (α, β, γ) of point P ?

$AB = AC = 5$ and $\angle CAP = 30^\circ$, $\angle PAB = 45^\circ$.

(Hint: $\text{area} = \frac{1}{2}ab \sin \theta$)



(4)

Answer:

$$\beta = \frac{\text{area}(PCA)}{\text{area}(ABC)} = \frac{\frac{1}{2} \cdot 5 \cdot 2 \cdot \sin 30^\circ}{\frac{1}{2} \cdot 5 \cdot 5 \cdot \sin 75^\circ} = \frac{\sqrt{6} - \sqrt{2}}{5} \approx 0.207$$

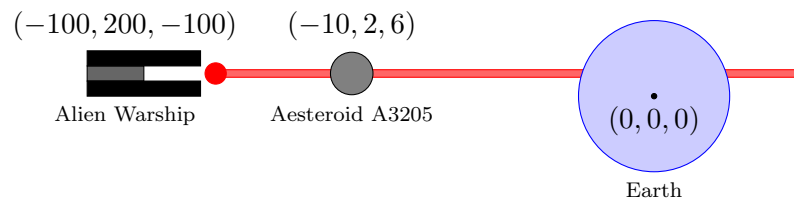
$$\gamma = \frac{\text{area}(PAB)}{\text{area}(ABC)} = \frac{\frac{1}{2} \cdot 5 \cdot 2 \cdot \sin 45^\circ}{\frac{1}{2} \cdot 5 \cdot 5 \cdot \sin 75^\circ} = \frac{2\sqrt{3} - 2}{5} \approx 0.293$$

$$\alpha = 1 - \beta - \gamma \approx 0.500$$

- b) If the vertices A , B and C are to be coloured red (255, 0, 0), green (0, 255, 0) and blue (0, 0, 255) respectively, what would be the interpolated colour of P in RGB? (2)

$$\begin{aligned} \text{Colour}(P) &= \alpha \cdot \text{Colour}(A) + \beta \cdot \text{Colour}(B) + \gamma \cdot \text{Colour}(C) \\ &= 0.500 \cdot (255, 0, 0) + 0.207 \cdot (0, 255, 0) + 0.293 \cdot (0, 0, 255) \\ &= (127.5, 52.785, 74.715) \\ &= \text{rgb}(128, 53, 75) \end{aligned}$$

3. a) A highly aggressive alien race shot their laser canon towards earth. It already destroyed the aesteroid A3205 in the solar system. Will earth survive? Radius of earth is 6400 KM. All units in 10^6 meters.



(4)

Answer: The ray origin is the alien warship's position $\mathbf{R}_o = (-100, 200, -100)$ and the ray direction is towards the aesteroid $(-10, 2, 6)$. We can find the direction vector $\mathbf{R}_d$ by subtracting the origin from the aesteroid's position and normalizing it.

$$\mathbf{R}_o = (-100, 200, -100)$$

$$\mathbf{R}_d = \frac{(-10, 2, 6) - (-100, 200, -100)}{|(-10, 2, 6) - (-100, 200, -100)|} = \frac{(90, -198, 106)}{\sqrt{58540}}$$

We normalize the direction vector to use the vector approach.

Step 1: Check if origin $\mathbf{R}_o$ is inside earth.

$$|\mathbf{R}_o| = \sqrt{(-100)^2 + 200^2 + (-100)^2} = \sqrt{60000} \approx 244.95 > 6.4$$

Step 2: Find the distance to the closest point on the ray to the origin of earth.

$$t_p = -\mathbf{R}_o \cdot \mathbf{R}_d = - \left((-100, 200, -100) \cdot \frac{(90, -198, 106)}{\sqrt{58540}} \right) \approx 244.6782$$

Step 3: Check that t_p is positive.

Step 4: Find the distance of the closest point on the ray to the origin of earth.

$$d = \sqrt{|\mathbf{R}_o|^2 - t_p^2} \approx \sqrt{60000 - 244.6782^2} \approx 11.5$$

Step 5: Check if $d < 6.4$. Since $11.5 > 6.4$, the ray does not intersect earth.

Therefore, earth survives.

- b) If earth survives, by how far did the laser miss? If not, at what distance from the Alien Warship did the laser hit earth? (2)

Answer: The distance missed the surface of earth by,

$$d - 6.4 \approx 11.5 - 6.4 = 5.1 \text{ (in } 10^6 \text{ meters)} = 5100 \text{ KM}$$

Which is still a close call! Maybe we should retaliate...

4. For your hard work and attending the exam so early in the morning. (2)

Answer: Thanks, I suppose.